

Nagarjun Mallesh

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SUMMARY

Results-driven GenAI Software Engineer with 4+ years designing distributed systems and cloud architecture solutions. Led cross-functional teams delivering 70% performance improvements through **AWS Bedrock** implementations and scalable solutions. Expertise in **system design**, **RAG pipelines**, **serverless architecture**, and **AI/ML** integration. Proven track record of reducing operational costs by 60% while maintaining 99.9% uptime across production environments.

EXPERIENCE

Senior Systems Analyst | ML Technologies LLC, Boston, MA | Jul 2025 - Present

- Optimized **emergency response** operations by implementing **GenAI** vehicle recovery system with **AWS Bedrock** and **OpenSearch vector search**, delivering 70% faster response times and processing 5K+ daily emergency requests across 12 metropolitan areas
- Established technical roadmap and mentored three junior engineers on GenAI best practices, accelerating team delivery velocity by 40% while maintaining code quality standards
- Designed an emergency services platform with API Gateway/Lambda processing 150K+ monthly requests with 99.9% uptime

Backend Software Engineer | Keelworks Foundation, Oak Harbor, WA | Jul 2024 - Jul 2025

- Led cross-functional team of 4 engineers and **QA specialists** to deliver production **RAG chatbot** using **AWS Bedrock**, serving 10K+ volunteers with 45% reduction in **AI hallucinations** through **RAGAS evaluation framework** implementation
- Designed distributed systems architecture for a serverless platform handling events through EventBridge-Lambda workflows, delivering scalable solutions with 60% cost reduction
- Designed a blue-green deployment pipeline, reducing deployment time from 4 hours to 8 minutes with zero-downtime releases

Software Engineer | Hewlett-Packard Inc, Bangalore, India | Oct 2019 - Jul 2022

- Delivered global automation solution by developing **PowerShell BIOS** configuration system deployed across multiple teams, transforming 4-day manual processes into 15-minute automated workflows while eliminating 90% of configuration errors
- Built Management Platform:** Orchestrated cross-functional engineering and QA teams to architect React.js/Node.js dashboard
- Owned Infrastructure Project:** Led AWS-based OS deployment automation, reducing provisioning from days to hours, deployed to 100K+ client machines
- Resolved critical production issues affecting 1000+ deployments within 24-hour SLAs, solutions adopted globally

Associate Software Engineer | Digital API Craft Pvt Ltd, Bangalore, India | Jul 2019 - Oct 2019

- Executed microservices migration for banking system using Spring Boot/Kafka, handling 50K+ daily transactions with 99.8% reliability
- Built a domain provisioning system using Golang with DNS integration, automating client onboarding, and reducing errors by 90%

SKILLS

Programming Languages: Java, Python, JavaScript/TypeScript, Node.js, Golang, PowerShell, C#

Cloud Platforms: AWS (Bedrock, Lambda, API Gateway, EventBridge, S3, EC2, CloudFormation), GCP, Azure, Docker, Kubernetes

AI/ML Technologies: Machine Learning Algorithms, RAGAS Framework, LangChain, RAG Pipelines, OpenAI APIs, Vector Databases

Data Technologies: MongoDB, PostgreSQL, Redis, Apache Kafka, OpenSearch, DynamoDB

Infrastructure: Terraform, Pulumi, CI/CD Pipelines, Serverless Architecture, Microservices

EDUCATION

MS Information Systems | *Northeastern University, Boston, MA* | 2022-2024

BE Information Science & Engineering | *Visvesvaraya Technological University, India* | 2015-2019

NOTABLE PROJECTS

- Multi-Agent Travel Planning System:** Engineered **AWS Bedrock-powered** platform leveraging **Claude/Nova** models with **API Gateway** deployment, reducing travel planning time by 80% and serving 2K+ monthly active users with a 95% satisfaction rating
- Concurrent Text Normalization Engine:** Engineered a multi-threaded system with AWS Bedrock LLM processing 10,000+ documents simultaneously with 95% accuracy
- AI Travel Matching Platform:** SpringBoot application with Binary Search Tree algorithms and OpenAI integration, serving 500+ users with 85% satisfaction